

School of Physics and Astronomy
PHYS5036 Detection and Analysis of Ionising Radiation

Experiment Information Script

Analysis of Dose Distributions in Radiotherapy

Understanding radiation detection requires a deep knowledge of how particles interact with matter. An alternative approach for studying particle interactions compared with the traditional use of detectors in the laboratory is to study via Monte Carlo simulation how different particle beams can be used for radiotherapy applications. The aim of this virtual experiment is to investigate how beams of different particles types and different energies deposit energy in a simulated water phantom.

Objectives:

- (i) To gain an understanding of the electromagnetic interactions of high-energy gamma-rays, electrons, protons and ions and relate the range of these particle types in matter to their underlying interactions.
- (ii) To understand the relationship between deposited energy, absorbed dose and equivalent dose.
- (iii) To become familiar with secondary particle interactions and hadronic processes and directly relate them to radiotherapy applications.
- (iv) To gain an appreciation of the differences in dose distributions in targeting cancer with different beam types.

Radiotherapy for Cancer Treatment

More and more treatment plans for cancer patients involve the use of radiotherapy, in which a beam of high-energy subatomic particles is used to cause catastrophic cell damage in cancerous tumours. The damage is done at the DNA level through microscopic interactions in which the beam particles transfer energy to the tumour tissue causing bonds to be broken and eventually cellular death. In order that the dose to which the tumour is exposed is much larger than the dose received by healthy tissue, treatment plans normally involve the use of shaped beams targeting the tumour area from several different orientations as shown in Figure 1 (taken from <http://www.aafp.org/afp/2008/1201/p1254.html>).

While the effectiveness of such treatment has been demonstrated in the clinical setting, there is a large international research effort into using different types of beams. Much of the motivation for this research is due to the need to further minimise the damage caused to healthy tissue during these types of treatments. There has therefore been a paradigm shift away from the use of high-energy gamma-ray beams to the use of hadron (proton or heavy ion) beams. Indeed, the NHS in England is in the process of developing two new hadron therapy treatment facilities. The shift from gamma to hadron beams is intimately connected to the nature in which these different particles interact and deposit dose in the patient. The motivation for the current virtual experiment is exploring the relevant particle interactions in this radiotherapy context.

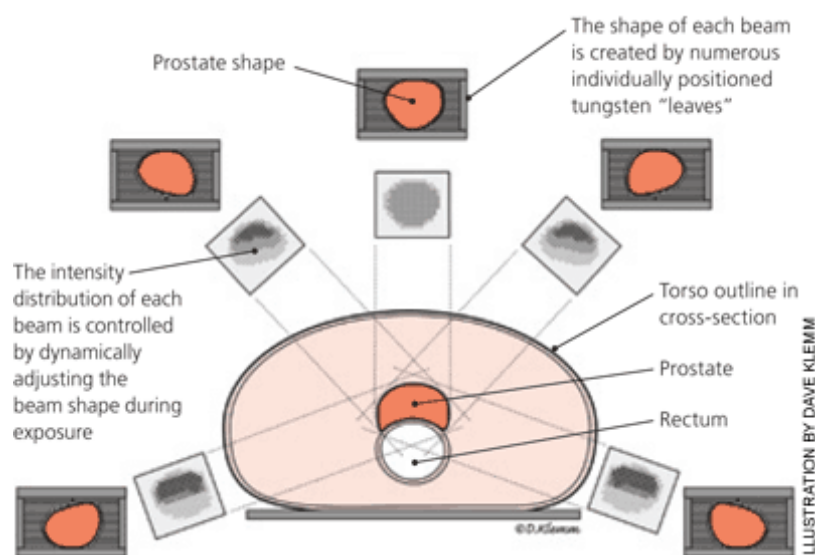


Figure 1: An example showing the use of several beam shapes, orientations and intensities

Software Installation Instructions

Before we can start investigating radiotherapy with different beam types, the software must first be downloaded, set up correctly and compiled on the School's UNIX teaching server.

First follow the steps in the "First Computing Set Up Steps" document, found on the Lab Moodle page, under the Lab Material → Computing section. It provides some useful introductory reading for using the School's computing.

Once you are happy with that document, log into JupyterHub in the normal way (remember if you are working remotely, you will need to first connect via VPN).

<https://jupyter.physics.gla.ac.uk/hub/login>

PHYS5036 Detection and Analysis of Ionising Radiation Laboratory

Then follow the environment set up instructions in Section 2 of the “Experiment Computing Set Up” document from on the Lab Moodle page, under the Lab Material → Computing section.

If you have any questions during this setup stage speak to the lab staff, who are happy to help you.

When we execute the command `bash /local/cern/software/Sim/SoftwareSetup.sh` during the Experiment Computing Set Up, we are telling your Unix environment how to find the relevant simulation code. This is a Geant4 particle transport Monte Carlo simulation called DoseProfileMC which has been designed specifically for this laboratory.

Advanced: For students interested in the Geant4 software details, the C++ class definition files can be found on Moodle in the Computing section, in a folder called “DoseProfileMC Source Code.” Have a look at the folders and files inside. The class implementation files are in `DoseProfileMC/src/` subdirectory. Have a look through these classes if you are curious how the C++ code for a Geant4 simulation is run and ask if you have any questions. User control of the simulation can be implemented by editing text files like those found in the `macros/` subdirectory: `DetectorSetup.mac`, `vis.mac` and `batch.mac`. Have a look at the commands in these files if you are interested in how Geant4 is run.

Electromagnetic Interactions of Gamma-rays

The energy-loss process for high-energy gamma-rays or charged particles is dominated by electromagnetic interactions between the incident particle and atomic electrons in matter. The nature of these interactions is dependent on particle type (charge and mass) and energy. In radiotherapy, it is these electromagnetic interactions that play by far the largest part in governing the nature in which energy is transferred from the beam particle to the patient.

We will investigate the energy dependence of the electromagnetic interactions for the four different beam types of interest (gamma-rays, electrons, protons and ions) by simulating a beam incident on a water phantom.

To begin with we will generate a beam of 1 MeV gamma-rays. This is a good starting point as radiotherapy is currently dominated globally by treatment plans and technologies involving high energy gamma-rays.

We can simulate this by running the following commands in a UNIX terminal. First we will make a folder called data to keep the simulation data in.

```
mkdir ~/data;  
cd ~/data;
```

```
runDoseProfile.py [beamtype] [energy in MeV] [N Particles] [Tumour Flag] [Hadronic Flag]
```

Where, we will want to set:

[beamtype] = proton

[energy in MeV] = 1

[N Particles] = 10 000 (feel free to play around with this to study the effect of statistics, but let's start with 10k)

[Tumour Flag] = 0, since we do not want to simulate a tumour yet

[Hadronic Flag] = 0, since at the moment we are only going to use electromagnetic physics libraries in Geant4

So the command you should actually enter in the terminal is:

```
runDoseProfile.py gamma 1 10000 0 0
```

PHYS5036 Detection and Analysis of Ionising Radiation Laboratory

This executes a python script (stored elsewhere on the JupyterHub system, at /local/cern), which passes the beam and physics set up parameters to the Geant4 simulation.

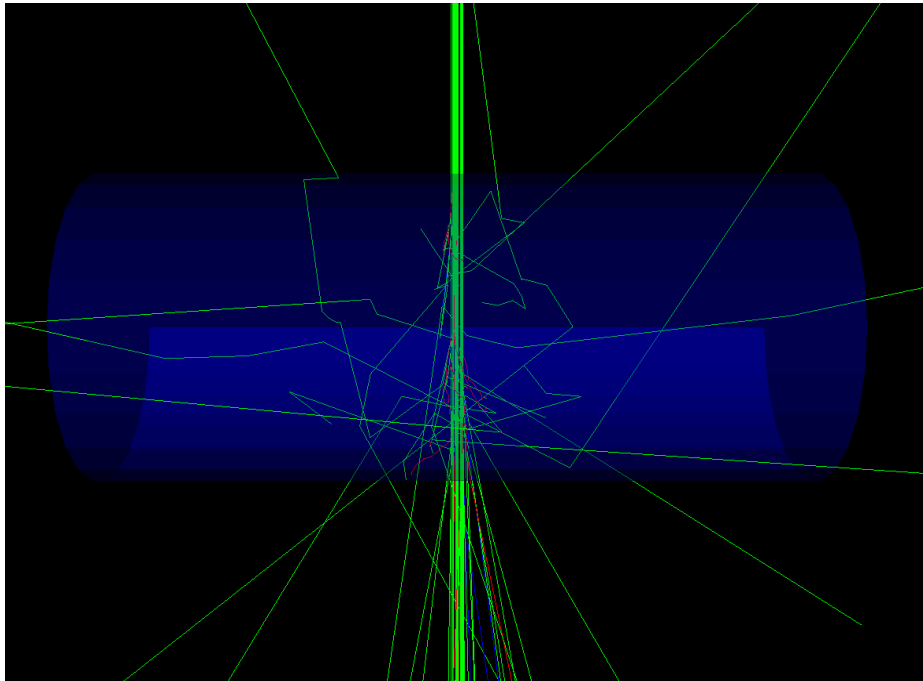


Figure 2: Example of a visualisation of the simulation

Figure 2 shows what you would see in a visualisation of the simulation: the blue cylinder is the water phantom with dimensions representative of a typical patient. The green, blue and red lines represent the tracks of different particle types (by charge) resulting from the gamma beam.

We are not running the visualisation ourselves, since Geant4 does not have this capability with a JupyterHub interface yet unfortunately, so do not expect to see this on your screen and please use the image above instead to answer the following question. What observations can already be made about the nature of the interaction between the beam and the phantom?

It is of course important to be more rigorous in these investigations than simply visualising the interactions of a handful of gamma-rays. We do this by investigating in detail how energy is deposited in the phantom by analysing the simulated data in root.

To start your analysis navigate to the python script in your ~/data directory with your mouse (in the file browser on the left hand side of the JupyterHub launcher). Open ~/data/DoseProfileAnalysis.ipynb. You can modify and expand this notebook however you wish. Have a read through the notebook and a look at the raw data output from the simulation.

The output from the simulation is recorded in a csv file (you should also see this in your ~/data folder with filename DoseMC_proton_1MeV_10000n_Out.csv). Opening the csv file may be slow, since it is a lot of data. In the simulation, the phantom is split up into cubic volume elements (voxels). The energy deposited in each voxel is recorded, as well as the type of particle(s) which deposited the energy in each voxel. This information is stored in the csv file. Ignore the columns referring to “T” for the moment, these will become apparent later in the script.

After reading in the csv data into a dataframe, this notebook performs three main analysis plots of the data.

PHYS5036 Detection and Analysis of Ionising Radiation Laboratory

- Dose profile by particle type
- Longitudinal dose profile (i.e. parallel to the incident beam direction)
- Transverse Dose Profile (i.e. transverse to the incident beam direction)

The dose profile by particle type created a histogram (frequency distribution) of deposited energy for different particle types. The particle type in this case refers to the particle which deposited the energy and can be either lepton, meson, baryon or ion (in this case, only leptons are relevant). It also prints out the energies deposited by the particles. The longitudinal and transverse plots are similar in that they plot the distribution of energy deposit in a central slice through the phantom in the longitudinal (parallel to the beam) and transverse (perpendicular to the beam) direction, respectively.

Look now at the results for the 1 MeV gamma-rays. What can you say about the magnitude and distribution of energy deposited in the phantom for a beam of this type? How do your observations relate to the underlying gamma-ray interactions?

Many radiotherapy machines that are in use for treatment around the world (especially in developing countries) operate with similar beams to the one we just simulated. The source of gamma-rays for these devices is a rod of Co-60 (with gamma-ray energies of 1.173 and 1.333 MeV). In the UK radiotherapy technology is based on the use of compact electron linear accelerators producing higher energy gamma-rays beams via bremsstrahlung.

By investigating how the energy transferred to the phantom varies with incident gamma-ray energy (i.e. running the simulation for beams with kinetic energies of 5, 10 and 50 MeV), what conclusions can be made in terms of the old and new technologies? How do the nature of the underlying interactions change as the beam energy is increased? What other factors must be considered when thinking about the advantages and limitations of the accelerator-based technology.

Deposited Energy, Absorbed Dose and Equivalent Dose

Up to now, when discussing interactions between the beam and the phantom we have been interchanging the terms deposited energy and dose. In order to understand, in a quantitative sense, the efficacy of a particular radiotherapy technology it is important to define formally what is meant by these terms and the relationship between them.

The stochastic effect in terms of cell damage as a result of exposure to ionising radiation is characterised by the equivalent dose H measured in Sieverts (Sv). The equivalent dose absorbed in tissue is defined such that:

$$H = \sum_R W_R D$$

where W_R is a weighting factor for radiation type R , which is a measure of the relative biological effectiveness of the radiation type. Nominal values of W_R for different particle types as set by the International Committee for Radiation Protection (ICRP) are:

Radiation Type W_R

lepton	1.0
meson	2.0
baryon	2.0
ion	20.0

PHYS5036 Detection and Analysis of Ionising Radiation Laboratory

The absorbed dose D is defined as the mean energy deposited by ionising radiation in matter (E_d) per unit mass, the unit of which is the Gray (J/kg):

$$D = E_d / m$$

The simulation records the energy deposited in a given volume element (or voxel) of the phantom. The header variable of the csv files gives the voxel dimensions (where min and max variables are in mm).

For the longitudinal dose profile, voxels are combined in such a way to create individual slices of volume lying along the beam direction and the mean energy per slice is output. Using the Jupyter notebook, calculate the volume of each slice used in the longitudinal dose profile calculation which gives the result called mean energy deposited per voxel (note that the word “voxel” is used rather than “slice” in the notebook for the results output in the dose profile calculations). Note that the slice calculation uses filterframe not cutframeL. Using these numbers calculate the total absorbed dose (D) and equivalent dose (H) per incident beam particle for the four gamma-ray energies you just simulated.

How many 10 MeV gamma-rays would a patient need to be exposed to achieve a absorbed dose of 50 Gy in this volume?

Looking at the 2D distribution of x-z energy deposits, estimate a rectangle which contains the core of the energy deposited by the beam. Create a new cell in your notebook which filters on the dataframe to only select these voxels and use this to recalculate D and H . e.g.

```
newframe = filterframe[(abs(filterframe['zpos'])<10) & (abs(filterframe['xpos'])<10)] “
```

Please note you should modify your python script to perform these calculations.

Electromagnetic Interactions of Heavy Charged Particles

Electromagnetic energy loss for heavy charged particles is dominated by the ionisation process as described by the famous Bethe-Bloch equation, in which the ionisation density is related to the charge, mass and velocity of the incident particle. As the particle slows, the cross section for interactions with atomic electrons increases leading to higher and higher energy loss until the particle comes to rest. It is this behaviour, characterised by the presence of a Bragg peak in the dose distribution for heavy charged particles, that has led to intense interest in proton and ion radiotherapy in recent years.

In order to demonstrate the significant difference between gamma-ray and hadron radiotherapy, we are going to run the simulation with beams of protons and carbon ions. As before, let's simulate and extract the dose distributions for proton beams with kinetic energies of 100, 500 and 1000 MeV. Now let's do the same for ion beams of 500, 1000 and 2000 MeV.

By looking at the resultant plots, what can you conclude about the nature of the energy deposit for hadron beams? What can you say about the range of the charged particles in the water phantom, and how might this be exploited in cancer treatment? How does the longitudinal and transverse dose profile differ from the gamma-ray data? Comment on the typical length scale (i.e. FWHM) associated with the Bragg peaks for protons and ions of the same energy. What implications does this latter observation have in terms of treatment planning?

Realistic Radiotherapy Simulations with Hadronic Interactions

PHYS5036 Detection and Analysis of Ionising Radiation Laboratory

The investigations up to now have focussed on electromagnetic interactions between the beam particles and the phantom. In order to fully understand the dose profiles in radiotherapy applications, we must also include in the simulation hadronic interactions between the beam and phantom. Take a short time to look into what is meant by hadronic interactions and how they lead to a higher dose being deposited in the phantom.

Before including these interactions, we are first going to make the radiotherapy simulation more realistic by placing a small sphere (1 cm diameter) at the centre of the phantom. By recording the energy deposited in this sphere separately from the rest of the phantom, we can start to quantify the effectiveness of different therapy treatments. Note this corresponds to the columns with a “T” identifier in the csv file. To quantify the effectiveness, we consider the sphere to be the tumour that is being targeted and the rest of the phantom to represent healthy tissue. This then allows us to simulate and measure the Tumour-to-Normal (TNR) ratio, which as the name suggests is simply:

$$\text{TNR} = D_{\text{tumour}} / D_{\text{healthy}}$$

The ultimate goal in radiotherapy treatment planning is keeping TNR as high as possible in order to minimise the damage sustained in the patient's normal or healthy tissue.

Now re-run your previous simulations to include the spherical tumour volume, setting the [TumourFlag]=1, e.g.

```
runDoseProfile.py gamma 1 10000 1 0
```

Next, use the data to identify the optimum energy for targeting this tumour (i.e. maximising the TNR) for the three different beam types. This will require a bit of trial and error. You are not restricted to the energies suggested in this script and can fine tune the values until you get an optimum result. Once you are happy that you have optimised the beam energies, run the simulations again and take note of the results. Pay particular attention to the fraction of energy deposited by different particle types.

Finally, add hadronic interactions into your simulation by setting [Hadronic Flag]=1, i.e. including hadronic interactions in our simulation.

```
runDoseProfile.py gamma 1 10000 1 1
```

This performs the following change to the physics libraries used in Geant4:

```
from: /DoseProfileMC/physics/addPhysics standard_opt3  
to: /DoseProfileMC/physics/addPhysics QGSP_BIC_EMY
```

You can search online for more information on these libraries if you are interested.

Re-run your three optimised beam scenarios. What can you say about the difference in dose distribution with and without hadronic interactions? Which beam type is affected most by this change and why? The addition of hadronic interactions means that you should see more secondary particle production. How do these secondary particles contribute to the dose magnitudes and distributions?